TalentScout

Football Academy Management & AI Player Development

MERN Stack Edition (MongoDB, Express.js, React, Node.js)

1. Project Overview

TalentScout is a football academy management platform designed to help coaches organize training activities, manage young players, and monitor their development over time.

Football academies regularly evaluate players based on physical fitness, technical abilities, discipline, and training performance. However, managing this information through spreadsheets, paper records, or disconnected tools can make it difficult to maintain consistent evaluations and understand how individual players are progressing.

TalentScout addresses this challenge by providing a centralized platform where academies can manage player profiles, organize training sessions, record attendance, and maintain structured performance evaluations.

The platform also integrates artificial intelligence to analyze performance data recorded by coaches. Based on this information, AI can generate individual progress summaries, highlight strengths and weaknesses, and suggest personalized training recommendations.

The project aims to achieve three main objectives:

- **Simplify academy management:** Provide coaches with practical tools to manage players, training sessions, attendance, and evaluations.
- **Monitor player development:** Maintain a structured history of individual performance that allows coaches to observe improvements and identify areas requiring attention.
- **Support coaching decisions with AI:** Transform recorded performance data into useful insights and personalized training recommendations.

The final application should provide a complete workflow, from player registration and training organization to performance evaluation and AI-assisted development analysis.

2. Users and Roles

TalentScout supports two main user roles.

Academy Manager

The Academy Manager is responsible for organizing the academy and overseeing its activities. The manager should be able to:

- Manage the academy's basic information.
- Create and manage coach accounts.
- Register and manage players.
- View training sessions and attendance records.
- Access player evaluations and development information.
- Monitor academy activities and overall player progress.

Coach

Coaches are responsible for organizing training activities and evaluating player development. They should be able to:

- View and manage players under their responsibility.
- Create and organize training sessions.
- Record player attendance.
- Evaluate players using relevant football performance metrics.
- Review previous evaluations and track individual progress.
- Generate AI-assisted performance summaries and training recommendations.

Both roles must have secure authentication and access permissions appropriate to their responsibilities.

Players are managed as academy records and do not require individual accounts in the core application.

3. Functional Requirements

3.1 Must Have: Required Features

The following features are essential for TalentScout to function as a complete football academy management and player development platform.

A. Authentication and User Management

The platform must provide secure authentication and role-based access control.

- Academy managers and coaches must be able to log in using email and password.
- Academy managers should be able to create, edit, and manage coach accounts.
- Users should be able to manage their basic profile information.
- Coaches should only access players and training information they are authorized to manage.
- The application must protect player information and restrict management actions based on user roles.

B. Player Management

TalentScout must provide a centralized system for managing academy players. The platform should allow authorized users to:

- Register new players with their personal and football-related information.
- Store relevant information such as name, date of birth, playing position, and registration date.
- Create, view, edit, and archive player profiles.
- Organize players into appropriate training groups or age categories.
- Assign players to coaches or training groups.
- Access each player's training attendance and performance history.

Each player should have a dedicated profile that brings together their general information, training participation, and development records.

Player records should remain available for historical analysis, even when a player is no longer actively training at the academy.

C. Training Session Management

Coaches must be able to organize and manage training sessions. The platform should allow coaches to:

- Create training sessions with a title, date, time, and relevant details.
- Specify the objectives or focus of a session, such as endurance, passing, ball control, or tactical awareness.
- Associate training sessions with specific groups or players.
- View upcoming and previous training sessions.
- Edit or cancel scheduled sessions.
- Record relevant session notes after training.

Training sessions should be connected to the academy's player management and attendance functionality.

D. Attendance Tracking

TalentScout must allow coaches to record and monitor player attendance.

- Coaches should be able to view the players expected to attend a training session.
- Each player's attendance should be recorded using appropriate statuses, such as Present, Absent, or Excused.
- Coaches should be able to update attendance records when necessary.
- Attendance history should be accessible from individual player profiles.
- The platform should provide basic attendance summaries for players and training sessions.

Attendance data should help coaches understand player participation and training consistency.

E. Player Performance Evaluation

Coaches must be able to evaluate players using structured football performance metrics. The application should support evaluations across several relevant categories.

Evaluation Category	Example Metrics
Physical Abilities	Speed, endurance, agility, strength
Technical Skills	Passing, shooting, dribbling, ball control
Tactical Abilities	Positioning, decision-making, game awareness
Personal Development	Discipline, teamwork, motivation

These metrics are examples. The application should use a consistent evaluation approach that allows meaningful comparisons over time.

The platform must allow coaches to:

- Create individual performance evaluations for players.
- Record scores or ratings for relevant performance metrics.
- Add written observations or comments.

- Associate evaluations with dates to maintain a historical record.
- View and update evaluation information.
- Access a player's previous evaluations.
- Compare current performance with previous results.

The evaluation process should make it possible to identify improvements, declining performance, or skills requiring additional attention.

Evaluations should be based on observations and measurements entered by coaches, not automatically extracted from videos or images.

F. AI-Assisted Player Progress Analysis

TalentScout must integrate AI to analyze recorded player performance information. Coaches should be able to request an AI analysis for an individual player. The application should:

- Collect relevant player information and available performance evaluations.
- Provide historical evaluation data to the AI service when available.
- Generate a readable summary of the player's current performance.
- Highlight demonstrated strengths and areas requiring improvement.
- Compare historical evaluations using consistent metrics and rating scales, identifying supported progress, regression, or stable performance only when sufficient comparable records exist.
- Suggest actionable, age-appropriate training activities that identify the targeted skill, specific exercises, objectives, suitable duration or repetitions when relevant, and the evaluation evidence supporting each suggestion.
- Allow coaches to access generated analyses within the player's development records.

AI recommendations should suit the player's age, position, documented abilities, and development needs. Coaches should be able to trace each recommendation to recorded evaluation results or written observations.

The analysis must support its conclusions with recorded evaluations and coach observations, distinguish evidence from general suggestions, and explicitly acknowledge insufficient or inconsistent data rather than inventing performance trends.

G. Player Development Overview

Each player should have a dedicated development overview that brings together important information. The overview should include:

- Basic player information and playing position.
- Training attendance history.
- Latest performance evaluation.
- Historical evaluation records.
- Available AI-generated progress summaries and recommendations.

Coaches should be able to understand a player's current development status without searching through disconnected records.

3.2 Should Have: Recommended Features

These features improve the academy's ability to monitor player development and make informed coaching decisions.

A. Performance Progress Charts

The application should provide visual representations of player performance over time. Coaches should be able to follow changes in selected metrics, such as speed, endurance, passing, or technical ability. Charts should make it easier to identify improvement trends and performance fluctuations.

B. Historical Evaluation Comparison

Coaches should be able to compare two or more evaluations for the same player. The comparison should highlight:

- Metrics that improved.
- Metrics that declined.
- Areas showing little or no change.
- Relevant differences between evaluation periods.

C. Individual Development Goals

Coaches should be able to define development goals for players. For example, a coach could set a goal to improve passing accuracy or endurance over a specific period. Each goal should include a target, an expected timeframe, and a progress status. Development goals should be connected to the player's performance evaluations whenever possible.

D. Academy Performance Overview

Academy managers should have access to an overview of academy activity, including:

- Number of active players and coaches.
- Upcoming and completed training sessions.
- General attendance statistics.
- Recent player evaluations.
- Players whose evaluations have not been updated recently.

3.3 Could Have: Optional Features

The following improvements encourage creativity and additional technical exploration.

- **Training Groups and Categories:** Provide more advanced organization of players by age, position, or development level.
- **Player Development Reports:** Generate downloadable reports summarizing individual player progress and recommendations.
- **AI Training Plans:** Generate structured training plans for individual players based on their evaluations and development goals.
- **Attendance Insights:** Identify recurring absence patterns and their possible relationship with training consistency.

- **Player Comparison:** Compare players using selected performance metrics while accounting for relevant differences such as age and playing position.

Students are encouraged to propose additional features that strengthen academy organization, player development, or coaching effectiveness.

4. AI Integration

4.1 Purpose

The primary purpose of AI in TalentScout is to help coaches interpret player performance information and make more informed development decisions.

Coaches regularly record evaluations, but understanding long-term progress across multiple metrics can require considerable analysis. TalentScout uses AI to transform these records into understandable progress summaries and practical training recommendations.

AI should assist coaches in interpreting recorded performance, rather than replacing their professional judgment.

4.2 Expected AI Workflow

The AI functionality should support the following workflow:

- **1. Player Evaluation:** A coach evaluates a player by recording scores for physical, technical, and tactical abilities, along with relevant observations.
- **2. Performance History:** The platform stores the evaluation and maintains the player's previous performance records.
- **3. Analysis Request:** The coach requests an AI-generated progress analysis for the selected player.
- **4. Data Preparation:** The application provides the AI service with relevant player information, recent evaluations, historical results, and coach observations.
- **5. AI Analysis:** The AI compares evaluations using consistent metrics, checks recorded scores and coach observations for evidence, and identifies supported strengths or development trends. When the available history is insufficient, it explains the limitation.
- **6. Training Recommendations:** The AI suggests age-appropriate exercises with targeted skills, objectives, suitable durations or repetitions where relevant, and references to the evaluation findings supporting each recommendation.
- **7. Coach Review:** The coach reviews the analysis and uses the recommendations when planning future training activities.

The AI analysis should become more informative as additional performance evaluations are recorded.

4.3 Expected AI Capabilities

The AI integration should be capable of:

- Interpreting structured football performance metrics and coach observations.
- Summarizing a player's current performance.

- Identifying strong and weak areas based on recorded evaluations.
- Comparing historical evaluations using consistent metrics and scales, supported by recorded results and coach observations.
- Generating actionable, age-appropriate training recommendations with targeted skills, specific exercises, objectives, appropriate duration or repetitions, and evidence from evaluations.
- Considering a player's age category and playing position when relevant.
- Producing clear, professional summaries that coaches can understand and use.

For example, if a player consistently receives strong ratings in endurance but lower ratings in passing accuracy, the AI could highlight endurance as a strength and recommend passing-focused exercises.

If historical evaluations show improvement in passing, the analysis should recognize that progress rather than treating the latest score in isolation.

4.4 Technical Considerations

The AI functionality must be implemented within the Express.js backend, as a dedicated AI service module (for example, a separate layer that prepares the data, builds the prompt, calls the chosen AI API or model, and validates the response). No separate AI backend is required.

The AI analysis must rely on performance information entered by coaches.

Students should consider the following constraints:

- AI recommendations must be grounded in the player's recorded evaluations and observations.
- Missing, insufficient, or non-comparable evaluation data must be acknowledged. AI must not invent improvement, regression, scores, or trends.
- Historical comparisons must use consistent metrics and evaluation scales and should explain the recorded evidence behind conclusions, including relevant coach observations.
- Training suggestions must be age-appropriate, avoid unsafe workloads, and must not be presented as guaranteed performance improvements. AI must not diagnose injuries, assess medical conditions, or prescribe medical treatment.
- AI-generated summaries must remain understandable and relevant to the intended coaching context.
- Coaches should be able to review the results before applying recommendations. Access to player records, evaluations, and AI analyses must remain restricted to authorized academy managers and coaches.
- The solution should use free APIs, open-source models, or appropriate free-tier services.

Video analysis, automatic player detection, and computer vision are not required.

Students are free to select the most appropriate AI models, APIs, prompting strategies, and data-processing approaches.

5. Technology Stack and Deployment

5.1 Required Technology Stack (MERN)

The project must be built exclusively with the MERN stack. No other framework, language, or database is required or expected.

Component	Technology
Database	MongoDB (with Mongoose as the ODM)
Backend	Node.js with Express.js (main application logic and AI integration)
Frontend	React.js (functional components and hooks, with React Router for navigation)
AI Functionality	Player performance analysis and personalized training recommendations, integrated through the Express.js backend using an AI API or model
Version Control	Git and GitHub

MongoDB is well suited to TalentScout, particularly because evaluation records can be stored flexibly (for example, a category-based structure of metrics and scores) while players, coaches, training sessions, and attendance records are linked through references. Students should explain their data modeling choices; the quality of the reasoning and implementation matters.

The Express.js backend is responsible for the main application logic, authentication, authorization, database operations, and AI-related analysis. The React frontend consumes the backend REST API.

Students are responsible for selecting appropriate application architecture, database design, evaluation structures, and the organization of the AI module within the backend.

5.2 Deployment

The completed TalentScout application must be deployed and accessible online. All components, including the frontend, backend, database, and AI integration, must communicate correctly in the deployed environment.

Suggested deployment services include:

Component	Suggested Service
React Frontend	Vercel or Netlify
Express.js Backend (including AI integration)	Railway or Render
MongoDB Database	MongoDB Atlas

These providers are suggestions rather than mandatory choices. Students may choose alternative hosting services that satisfy the project requirements, preferably using available free tiers.

6. Expected Deliverables

The final project submission should include:

- A functional full-stack TalentScout application built with the MERN stack.
- Authentication and role-based authorization for academy managers and coaches.
- Coach account management.
- Player registration and profile management.

- Training session organization and attendance tracking.
- Structured player performance evaluations with historical records.
- AI-assisted player progress analysis and training recommendations.
- Individual player development overviews.
- Persistent database integration using MongoDB.
- Source code available in a GitHub repository.
- A README containing the project overview, installation steps, configuration, and deployment instructions.
- Sample coaches, players, training sessions, attendance records, and performance evaluations for demonstration.
- A live deployed application accessible online.
- A final presentation and demonstration of the application's main workflows and technical decisions.

Example End-to-End Scenario

- **1. Account Setup & Registration:** A football academy manager creates coach accounts and registers several young players in the academy.
- **2. Training Session Planning:** A coach organizes a training session for an under-15 group, focusing on passing accuracy, endurance, and ball control.
- **3. Attendance Tracking:** On the day of training, the coach records each player's attendance.
- **4. Initial Player Evaluation:** After the session, the coach evaluates a player named Adam, a 14-year-old midfielder, by assigning performance ratings for passing, endurance, agility, and tactical awareness.
- **5. Observation Logging:** The coach also adds a note indicating that Adam demonstrates good stamina but sometimes struggles with accurate passes under pressure.
- **6. Follow-up Evaluation & AI Request:** Several weeks later, the coach records another evaluation and requests an AI-generated progress analysis.
- **7. AI Progress Analysis & Recommendations:** The AI compares Adam's evaluations using the same rating scales and cites the recorded results and the coach's observation about inaccurate passes under pressure. It identifies improvements only where the data supports them and recommends age-appropriate passing drills with a clear objective, suitable repetitions, and an explanation tied to his evaluations.
- **8. Coach Review & Application:** The coach reviews the recommendations and uses them to guide Adam's future training.
- **9. Manager Oversight:** The academy manager can access the player's development history and review his attendance and performance records.

This scenario demonstrates the expected connection between player management, training organization, attendance, performance evaluation, and AI-assisted development analysis.

7. Evaluation Criteria

The project will be evaluated according to the following criteria:

Criterion	Description
Functional Completeness	Correct implementation of the required features and main academy management workflows.
Frontend and User Experience	Interface quality, responsiveness, ease of use, and suitability for academy managers and coaches.
Backend and Database	Quality of APIs, data integrity, authentication, authorization, and business logic; sound MongoDB data modeling, justification, and implementation.
AI Integration	Evidence-based historical comparisons, appropriate handling of insufficient data, and practical, age-appropriate training recommendations.
Code Quality	Organization, readability, maintainability, and appropriate technical decisions.
Deployment	Accessibility and correct integration of all application components.
Team Collaboration	Effective collaboration, Git usage, individual contributions, and understanding of the implemented solution.
Final Demonstration	Ability to demonstrate complete workflows, explain technical choices, and present a functional application.

The expected outcome is a coherent, functional, and practical football academy management platform that demonstrates the ability to combine full-stack MERN development with artificial intelligence to support player development and coaching decisions.